

CSE 4810 : Algorithm Engineering Lab

Lab - 2

MD ATIQUR RAHMAN

Group - SWE A

June 16, 2025

Task 1

Sort the given list of integers (every integer will be unique) utilizing only tree traversals.

Input: 1 4 7 5 3 2

Output: 1 2 3 4 5 7

Task 2

Write a Python program to check whether a given binary tree is a valid binary search tree (BST) or not.

Let a binary search tree (BST) is defined as follows: The left subtree of a node contains only nodes with keys less than the node's key. The right subtree of a node contains only nodes with keys greater than the node's key. Both the left and right subtrees must also be binary search trees.

Input:

1: -,4

2: -,-

3: 2,-

4: 3,7

5: -,-

7: 5,-

Output: True

Input:

1: 2,3

2: -,-

3: -,-

Output: False

Helping Code

Implementation of creating a BST

```
# Python program to demonstrate  
# insert operation in binary search tree
```

```

class Node:
    def __init__(self, key):
        self.left = None
        self.right = None
        self.val = key

# A utility function to insert
# a new node with the given key
def insert(root, key):
    if root is None:
        return Node(key)
    if root.val == key:
        return root
    if root.val < key:
        root.right = insert(root.right, key)
    else:
        root.left = insert(root.left, key)
    return root

```